

CHURNIKA MARAPPA REDDY

Data Engineer | Machine Learning Engineer | Software Engineer

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Professional Summary

Graduate Data Science student (MS, Dec 2026) focused on data engineering and machine learning. Built end-to-end pipelines with Airflow, Spark and Iceberg, served ML models through FastAPI, and adapted diffusion-based generative models in research. Comfortable across Python, SQL, PostgreSQL, AWS and Docker, from raw data to a usable API or dashboard.

Technical Skills

Languages: Python, SQL, JavaScript, TypeScript, Java, C++, HTML, CSS

Data Engineering/Analytics: Airflow, Spark, Iceberg, ETL/ELT, PostgreSQL, Pandas, NumPy, Databricks, Data Validation, Feature Engineering

AI/ML: PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, MLflow, Optuna, LLMs, Generative AI, Computer Vision

Full Stack/DevOps: React, React Native, Next.js, FastAPI, REST APIs, Supabase, SQLite, Docker, Git, GitHub Actions, Linux, AWS S3

Experience

University at Buffalo, SUNY

Research Assistant - Generative Handwriting & Document Recognition

Buffalo, NY

Aug 2025 - Present

- Adapted DiffusionPen, a few-shot handwriting style-transfer system (latent U-Net, DDIM sampling, Stable Diffusion VAE, CANINE text and MobileNetV2 style embeddings).
- Engineered IAM and GNHK data and training pipelines with writer-identity mappings, cached tensors, dataset loaders, EMA checkpoints and style-encoder training.
- Built an end-to-end Streamlit handwriting-generation app with uploaded style images, CPU/GPU checkpoint loading, multi-candidate generation and style-based reranking.
- Evaluated Emuru (VAE + autoregressive Transformer) against the diffusion baseline and analyzed style fidelity on unseen writers.

Tirumala Tirupati Devasthanams

Web Development Intern

Tirupati, India

Jun 2024 - Aug 2024

- Debugged frontend, backend and API issues across enterprise web applications, tracing root causes with development teams and verifying fixes before release.

Slash Mark

Python Intern

Remote

Oct 2023 - Feb 2024

- Developed reusable Python programs for automation, data processing, analysis, debugging and algorithmic problem-solving using Git workflows.

Selected Projects

Citi Bike Demand Prediction Pipeline

Python, Airflow, Spark, PostgreSQL, ML

- Engineered a Bronze-Silver-Gold pipeline using Airflow, Spark, Iceberg, PostgreSQL, FastAPI and Docker for station-level demand prediction; achieved **50.9% lower cumulative operational penalty** than the naïve baseline in the documented evaluation window.

Evently - Mobile Event Discovery Platform

React Native, TypeScript, Supabase

- Built 10+ user flows including discovery, search/filtering, recommendations, communities, saved events, RSVP/ticketing, event creation and profiles.

Telco Customer Churn ML System

Python, Scikit-learn, XGBoost, MLflow

- Compared Logistic Regression, Random Forest, SVC and XGBoost across 16 tracked configurations with FastAPI/Streamlit serving.

Real-Time Weather Application

React, FastAPI, SQLite, REST APIs

- Full-stack app on live weather APIs with location search, geolocation, five-day forecasts, CRUD persistence, validation and JSON/CSV exports.

Cognitive Stress Identification

Python, TensorFlow, OpenCV

- Built a real-time computer vision pipeline combining emotion recognition, eyebrow displacement and blink-pattern analysis for multi-signal stress assessment.

Education

University at Buffalo, SUNY

M.S. in Engineering Science - Data Science

Buffalo, NY

Expected Dec 2026

Amrita Vishwa Vidyapeetham

B.Tech. in Computer Science and Engineering

Coimbatore, India

May 2025